Physics Concepts as Indicative Framework to Model Systems of Economies Ex-ante

Coincise Notes and Conjectures Explorations of Natural Laws as Approach to Preliminarly Structure Economies - A Distillation of Corroboration of Mathematical Models

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An exploratory attempt … Interconnection network between movement, force, momentum, energy, and work. Describing systems with Lagrangian and Hamltonian when constraints and starting conditions exist. Modeling systems and optimisation with constraints and conditions. Interwoven system of macro- and microeconomie under constraints, limitations, and conditions.

Table of contents

WORKING NOTES DRAFT

POTENTIALS

Potential as “intrinsic energy of inertia”

Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := rot(A) \quad \overset{\text{Matter}}{=} \mu_0 \left(\mathcal{H} + \mathcal{M}\right) \overset{\text{Vacuum}}{=} \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \quad \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(c p_{0_{dipol}} + m_0 \tag{1}$$

Magnetic Field in Matter

Exitation $\widehat{=}$ Displacement

$$\text{Excication:} \qquad \mathcal{H} = \frac{1}{\mu_0}\mathcal{B} - \mathcal{M} = \frac{1}{\mu}\mathcal{B} \quad \text{Magnetisation:} \qquad \mathcal{M} = \frac{\mu - \mu_0}{\mu\mu_0}\mathcal{B} = \frac{\mu - \mu_0}{\mu_0}\mathcal{H} \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1-\frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \Bigg[\underbrace{\left(1-\frac{v^2}{c^2}\right)\frac{\frac{r}{|r|}-\frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|}\times\Big[\Big(\frac{r}{|r|}}{\ }}_{\text{accelerate}} \tag{3}$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)
dielectricum = non-conducting material
electricum = conductor medium

$$\text{Excication:} \qquad \mathcal{D} = \varepsilon_0\mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0\mathcal{E} + \varepsilon_0\chi\mathcal{E} = \varepsilon_0\left(1+\chi\right)\mathcal{E} = \quad \varepsilon_0 4\pi r^3 n \left\langle p_m \right\rangle = 4\pi r^3 n \tag{4}$$

Abbreviations

$\Phi =$ electric potential $\qquad A =$ magnetic potential $\qquad j = \nabla \times \mathcal{H} =$ elec. current density

$\mathcal{E} =$ Electric Field $\qquad \mathcal{B} =$ Magnetic Field $\qquad \nabla \cdot \mathcal{B} = 0 \qquad \mathcal{D} =$ Displacement density of dielectricum $\qquad \mathcal{M} =$

$\mathcal{P} =$ Polarisation $\qquad \mathcal{H} =$ magn. exitation, magn. field strength ("magn. displacement")

$n =$ spacial molecule density in volume

$\alpha = 4\pi r^3 = constant \qquad \chi =$ elec. susceptibility of dieelectricum $\qquad \kappa = \frac{\mu - \mu_0}{\mu_0} =$ magn. susceptibility

$\langle p_m \rangle =$ spacial average of magn. dipol moment $\qquad \langle \mathcal{E}_m \rangle =$ spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field $\quad \mu$ = magn. permeability $\quad \mu_0$ = magn. constant ...

ε = dielectricity constant $\quad \varepsilon_0 = ...$

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} = i\omega\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} = j - i\omega\varepsilon_0\mathcal{E} \tag{5}$$

relativistic, fast charge movement:

...

Potential derivation

$\Rightarrow$ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{6}$$

$\Rightarrow$ Potential caused by "ignition J" (inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\,dr' \tag{7}$$

elec. Potential structure

$J = -\frac{1}{\varepsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\varepsilon_0}\int \frac{\varrho}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}It \tag{8}$$

magn. Potential structure

$J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi}\int \frac{j}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\varepsilon_0}I = \frac{1}{c^2}\frac{1}{t}U_e \overset{\text{source in uniformal movement},v=const}{=} \tag{9}$$

Electric endogen energy

- Bond Energy
- Binding Energy

$E_{pot} = mc^2$

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{10}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{\left(mc^2\right)^2}{ne} = \frac{m}{q}mc^4 = \kappa mc^4 \tag{11}$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Exitation Energy

$E_{kin} = \frac{1}{2}mv^2 \mathrel{\hat{=}} \left(pc\right)^2 \mathrel{\hat{=}} h\nu$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2 \tag{12}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{13}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{14}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{15}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{16}$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda\frac{1}{T} \stackrel{vacuum}{=} kc \quad \stackrel{matter}{=} k\frac{c}{\eta} = kc\frac{\tan\beta}{\tan\alpha} \mathrel{\hat{=}} kc\left(\mathcal{E}_\beta^{-1}\mathcal{E}_\alpha\right)^{-1} = kc\frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} \mathrel{\hat{=}} \sqrt{\frac{k}{m}} \mathrel{\hat{=}} \frac{1}{\sqrt{LC}} \tag{17}$$

Phase:

$$\varphi = k\left(\vec{r} \cdot \vec{k}\right) - \omega t := ks - \omega t = ... \tag{18}$$

Phaseshift:

$$\Delta\varphi = \Delta k\left(\vec{r} \cdot \vec{\Delta k}\right) - \Delta\omega t := \Delta ks - \Delta\omega t = ... \tag{19}$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \mathrel{\hat{=}} \mathcal{E}_\beta^{-1}\mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = ... \tag{20}$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielctric madium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$... \tag{21}$$

Dielectricity:

$$\varepsilon(\omega) = ... \tag{22}$$

Permeability:

$$\mu(\omega) = ... \tag{23}$$

Conductivity:

$$\sigma(\omega) = ... \tag{24}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \tag{25}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{26}$$

Energy Balance

Energy Balance

$$\frac{\partial}{\partial t}\sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \tag{27}$$

Abbreviations:

w_i = engery densities (electric, magnetic, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectros $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

Mechanical

...

Electromagnectic energy transfer

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \tag{28}$$

$\varrho = \sigma\delta =$ charge density in interior

$\sigma =$ charge density at border (surface)

$\delta =$ Dirac delta-function

$j = \sigma\left(\mathcal{E} + v \times \mathcal{B}\right) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t}\mathcal{D} \hat{=}$ elecectric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity $= \left|\bar{\mathcal{S}}\right|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}\left(w_e + w_m\right) + \ \cdot \mathcal{S} + j \cdot \mathcal{E}\right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int \left(j \cdot \mathcal{E}\right) dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}\left(LI^2\right) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{29}$$

Abreviations:

$$\Delta \cdot := \text{ div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogenity,cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \qquad C = \text{Capacitance} \qquad Q = \text{charge (ion)} \qquad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector, radiation intensity}$$

$$D_S = \int \left(\ \cdot \mathcal{S}\right) dr = \int \text{div}(\mathcal{E}\times\mathcal{H})\, dr = \text{dissipation engergy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}} =$ skin depth (surface) of conductor $\qquad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho =$ radius (extension) of conductor

$\rho = kr =$ "radius of curvature k"

$\mu =$ permeability of conductor $\qquad \sigma =$ conductivity of conductor (dielectricum)

Thermal

...

Gravitational

...

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \tag{30}$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{31}$$

$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{32}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electic Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emgergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu\lambda = \frac{\lambda}{T}$
$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \qquad \nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index η emerges: $\varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Contiunuity of chrge/matter, no Refraction index, Reflection

Dissipation in Matter

no Contiunuity of chrge/matter with Dielectricum, Refraction merges, Reflection, Absorbtion

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

[1] Wirkungsquantum

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \tag{33}$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{34}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{35}$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}\sqrt{1 - \frac{C}{L}R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta k v = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \overset{EM}{=} \ldots = \sqrt{\frac{1}{LC}} \overset{QM}{=} \frac{1}{2}\frac{p^2}{m}\frac{1}{\hbar} = \frac{(\hbar\vec{k})^2}{2m} \tag{36}$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v}\frac{1}{\lambda} = \frac{\omega}{k}\frac{1}{\lambda} = \frac{\omega}{2\pi} \tag{37}$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar}\omega\mathcal{T} + \nu_0 \tag{38}$$

$$e = \text{electron charge} \qquad V_0 = \text{Opposing Potential } (I = 0) \tag{39}$$

Energy Level: $i = [1, n] \in \mathbb{N}$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \tag{40}$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurance of an event/state by space distance and time periodicity as well as movement speed and reach length, $\quad v^2 = v_{ph}v_{gr}, \quad \omega t = 2\pi\nu t = \lambda \neq 0 \quad$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \tag{41}$$

Rest state of source charge

Statics localization of charge. Only spacial demependeny, no time dependency.

Energy transport ... Information transmission ...

$v = 0 \qquad \dot{v} = 0$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v = const. \ll c \qquad \dot{v} = 0$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \approx c \qquad \dot{v} = 0$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport … Information transmission …

$v \neq 0 \qquad \dot{v} \neq 0$

Movement in Vacuum

$\eta = \sqrt{\frac{1}{\varepsilon_0 \mu_0}} = 1 \qquad 2\pi = k\lambda \qquad v_{ph} = c \qquad \omega = kc$

$\varepsilon_0 \mu_0 = \frac{1}{c^2}$

Continuity: $\nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$

Movement in Matter

$\eta = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0 \mu_0}} = \frac{c}{v_{Ph}} \neq 1 \qquad v_{ph} = \pm\sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \qquad \omega = kv_{ph} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} = kc\frac{\tan(\beta)}{\tan\alpha} = k\frac{c}{\eta}$

$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$r_0 \ll r \ll \lambda \qquad kr \ll 2\pi \qquad \frac{w_m}{w_e} \ll 1$

r is small, fast movement

Far Field

$r \gg \lambda \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1 \qquad \frac{w_m}{w_e} = 1$

r is big, slow movement

High Frequency

$\omega \gg \frac{2}{\mu\sigma r_0^2}$

λ is big, ν frequency i small, short time period

Low Frequency

$\omega \ll \frac{2}{\mu\sigma r_0^2}$

λ is small, ν frequency i big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global: ....

Local: ....

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1}\mathcal{D} = \mathcal{E}^{-1}(\mathcal{P} + \varepsilon_0\mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1}\mathcal{H} = \mathcal{B}^{-1}\left(\frac{1}{\mu_0}\mathcal{B} - \mathcal{M}\right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1+\omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta\frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \overset{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Operators

Operators

$\nabla := grad =$ Gradient (Gefälle, Neigung)

$\nabla \cdot := div =$ Divergence (Quelle)

$\nabla \times := rot =$ Rotation (Wirbelung, Curl)

$\nabla^2 =$ Lagrange Operatonr (Krümmung, Beschleunigung)

$\Delta \cdot := div(grad) =$ Laplace Operatorn

$\Delta :=$ Difference, shift

1. Cockshott WP (2011) Classical Econophysics. Routledge Advances in Experimental; Computable Economics

2. Wladyslaw W (2013) Macroeconometric Models, 1st ed. Springer

3. Lichtenegger K (2015) Schlüsselkonzepte zur Physik, 1st ed. Springer Spektrum

4. Padmanabhan T (2015) Sleeping Beauties in Theoretical Physics, 1st ed. Springer

5. Wachter A, Hoeber H (2005) Compendium of Theoretical Physics. Springer

6. Carter M (2001) Foundations of Mathematical Economics. The MIT Press

7. Michaelides PG (2025) 21 Equations that Shaped the World Economy, 1st ed. Palgrave Macmillan Cham

8. Pires da Cruz J (2025) Introduction to the Physics of the Economy and Finance, 1st ed. Springer Cham

9. Cass D, Shell K (2014) The Hamiltonian Approach to Dynamic Economics, 1st ed. Elsevier, Academic Press

10. Haven E, Khrennikov AI, Robinson T (2017) Quantum Methods in Social Science - A First Course. World Scientific, Publishing Europe Ltd